A CANONICAL SPECTRAL SEQUENCE OF DERIVED COMPLETE MODULES COMPUTING THE COMPLETION OF A FILTERED COMPLEX

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 26555-26567 (canonical tag 0BKD) and the counted proof is the proof environment at source lines 26569-26591, which immediately follows it and is its own proof.

Editorial note. The statement and proof use the notation of graded pieces and of the completion of a complex from the surrounding section; the completion functor is the one whose description is carried as a navigation stub.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

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Allowed standard prerequisites (editorial).

- Derived completion with respect to a finitely generated ideal is given by the derived Hom from a fixed bounded complex, and by the finite cohomological dimension statement that complex may be taken of finite projective dimension. invoked at proof lines 26569-26573. Replaces the completion functor by a Hom complex from a bounded complex of projectives, which is the object whose filtration is studied.
- The Hom complex from a bounded complex of projectives into a complex represents the derived Hom, and applying it to a filtered complex produces a filtered complex with the induced filtration. used at proof lines 26574-26581. Produces the complex representing the completion together with the filtration whose spectral sequence is the one claimed.
- The spectral sequence of a filtered complex has the stated first page and differential bidegree, and for a finite filtration on each term it is bounded and converges to the cohomology of the whole complex. invoked at proof lines 26581-26590. Gives the identification of the first page with the cohomology of the completion of the graded pieces and the convergence statement.

Source reference. Original lemma 091V: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26154–26154. The reference is invoked at proof line 26569 to write derived completion as a derived Hom from a fixed object; the construction of the Hom complex and its filtration are inside the counted span.

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Source reference. Original lemma 0AAJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26507–26507. The reference together with the amplitude statement gives finite projective dimension of the representing object, used at proof lines 26570-26573; everything built on it is inside the counted span. The lemma itself is the subject of a separate card of this lane with disjoint coordinates.

Source reference. Original lemma 0A5P: sources/stacks-more-algebra/pinned-original-source.tex, source lines 12817–12817. The reference is the second input to the conclusion that the representing object has finite projective dimension, cited at proof lines 26570-26572; it is a standard pointer and the use made of it is inside the counted span.

Source reference. Original section 012K: sources/stacks-chow/tags.txt, source lines 1402–1402. The reference points to the framework of spectral sequences of filtered complexes, cited at proof line 26582 to obtain the spectral sequence with the stated first page; it is a pointer to the standard construction and the filtration is built inside the counted span.

Source reference. Original lemma 012W: sources/stacks-chow/tags.txt, source lines 1414–1414. The reference is the convergence criterion applied at proof lines 26588-26590 after the filtration is seen to be finite on each term; the finiteness argument itself is inside the counted span.

Lemma 0.2. *Let A be a ring and let $I \subset A$ be a finitely generated ideal. Let $K^\bullet$ be a filtered complex of A-modules. There exists a canonical spectral sequence $(E_r, d_r)_{r\geq 1}$ of bigraded derived complete A-modules with d_r of bidegree $(r, -r+1)$ and with*

$$E_1^{p,q} = H^{p+q}((gr^p K^\bullet)^\wedge)$$

If the filtration on each K^n is finite, then the spectral sequence is bounded and converges to $H^*((K^\bullet)^\wedge)$.

Proof. By Lemma 091V we know that derived completion is given by $R\operatorname{Hom}_A(C, -)$ for some $C \in D^b(A)$. By Lemmas 0AAJ and 0A5P we see that C has finite projective dimension. Thus we may choose a bounded complex of projective modules $P^\bullet$ representing C. Then

$$M^\bullet = \operatorname{Hom}^\bullet(P^\bullet, K^\bullet)$$

is a complex of A-modules representing $(K^\bullet)^\wedge$. It comes with a filtration given by $F^p M^\bullet = \operatorname{Hom}^\bullet(P^\bullet, F^p K^\bullet)$. We see that $F^p M^\bullet$ represents $(F^p K^\bullet)^\wedge$ and hence $\operatorname{gr}^p M^\bullet$ represents $(\operatorname{gr} K^\bullet)^\wedge$. Thus we find our spectral sequence by taking the spectral sequence of the filtered complex $M^\bullet$, see Homology, Section 012K. If the filtration on each K^n is finite, then the filtration on each M^n is finite because $P^\bullet$ is a bounded complex. Hence the final statement follows from Homology, Lemma 012W. □